

Notes: Chiappori and Salanié (JPE, 2000)

Testing for Asymmetric Information in Insurance Markets

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1 Big picture

- **Question.** Does asymmetric information show up in competitive insurance markets in the simplest empirical way theory predicts: among observationally identical agents, do those who choose more comprehensive coverage also end up having more accidents?
- **Setting.** The paper studies the French automobile insurance market using a large insurer survey for 1989. The main application focuses on young drivers, where the data are rich and experience-rating complications are weakest.
- **Core challenge.** A positive correlation between insurance coverage and accidents is not automatically evidence of adverse selection. It could also reflect moral hazard, omitted observables, misspecified functional forms, local heterogeneity, or even differences between claims and true accidents.
- **Main conceptual contribution.** The paper proposes a very general test of asymmetric information based on *conditional dependence*. In a competitive insurance environment with exclusive contracts, asymmetric information implies that, within an observable risk class, agents choosing more coverage should be riskier ex post. Empirically, that means a positive correlation between contract choice and accident occurrence after conditioning on the insurer's observables.
- **Why the test is important.** The authors argue that this prediction is surprisingly robust. It does not depend much on the exact pricing rule, on very specific utility assumptions, or even on whether the hidden-information problem takes the form of adverse selection or ex ante moral hazard.
- **Main empirical design.** The paper estimates the relation between comprehensive coverage and accidents using three complementary procedures:
 - two separate probits with a generalized-residual independence test,
 - a bivariate probit that estimates the latent correlation directly,
 - a fully nonparametric cell-by-cell χ^2 independence procedure.
- **Main result.** Across all three procedures, the paper finds no evidence of asymmetric information in French automobile insurance once observables are adequately controlled for.
- **Young-driver result.** For beginners, the estimated dependence between coverage and accidents is essentially zero. The parametric and nonparametric tests all fail to reject conditional independence.
- **More-senior-driver result.** The same conclusion holds for somewhat more experienced drivers. So the paper also rejects the idea that drivers learn private information about their own risk faster than insurers do, at least over the short horizon available in the data.
- **Moral-hazard test.** The paper exploits a peculiarity of the French bonus/malus system: some young drivers start with an unusually favorable bonus coefficient of 0.5, effectively facing weaker accident incentives. This institutional feature raises coverage demand but is associated with *lower*, not higher, accident probability, so the data reject the moral-hazard interpretation as well.

- **Interpretation.** The natural reading is not that heterogeneity in risk is absent, but that insurers already observe enough relevant information that the residual private-information problem is small.
- **Broader takeaway.** The paper is a warning against overinterpreting positive correlations in insurance data. Misspecification can easily create spurious evidence of adverse selection. Negative findings, when based on rich conditioning and multiple methods, may actually be more credible than positive ones.

2 Data and measurement (key objects)

2.1 French automobile insurance as the application

French law requires every car to carry at least *responsabilité civile* (RC) insurance, which covers damage to other people and their vehicles. Insurers also offer broader coverage, most importantly *assurance tous risques* (TR), which also covers damage to the insured's own car or driver. The empirical coverage choice in the paper is therefore mainly the decision between minimum liability coverage and comprehensive insurance.

A distinctive feature of the French system is the legally regulated *bonus/malus* experience-rating schedule. If a driver begins the year with coefficient b_t :

- an accident raises next year's coefficient to $b_{t+1} = 1.25b_t$,
- no accident lowers it to $b_{t+1} = 0.95b_t$,
- and the coefficient is bounded between 0.5 and 3.5.

Young drivers can also be surcharged, but the surcharge must decline after accident-free years. This institutional structure is central both for the sample design and for the later moral-hazard test.

2.2 Orders of magnitude in the market

The paper gives a few useful magnitudes for interpretation:

- the average RC premium in the sample is about 3,000 francs,
- the average car is worth about 50,000 francs,
- the annual accident probability is about 10%,
- fewer than 2% of drivers have more than one accident in a year,
- and TR is at least twice as expensive as RC, with an even larger premium for expensive cars.

Interpretation. The move from RC to TR is economically large, so if asymmetric information were important, the coverage decision should be a natural place to detect it.

2.3 Data source and sample construction

The data come from a 1990 survey conducted by the French federation of insurers (FFSA), which covered 21 companies accounting for roughly 70% of the French automobile insurance market. The sampling rate was 1/20.

The raw files contain:

- 41 variables on 1,120,000 contracts,
- 25 variables on 120,000 accidents,
- all referring to calendar year 1989.

After cleaning and merging, the authors focus first on a sample of 20,716 “young” drivers — drivers who obtained their licenses in 1986, 1987, or 1988. This restriction is not cosmetic; it is the paper’s main way to avoid contamination from endogenous driving histories.

2.4 Why young drivers matter

The paper concentrates on young drivers for three reasons.

- **More homogeneous experience.** Mixing very new and very senior drivers would create heteroskedasticity and large differences in the residual distribution.
- **Cleaner conditioning set.** Insurers observe past driving records, and these records matter a lot for pricing. Restricting to drivers with little or no history avoids treating insurer information as if it were private information.
- **Less endogenous experience rating.** The bonus/malus coefficient summarizes past accidents, so it is potentially endogenous. Focusing on beginners is the cleanest way to sidestep this problem.

The sharpest beginner subsample used in the main parametric exercise is the set of drivers who obtained their license in 1988. There are 6,333 such individuals.

2.5 Coverage and accident variables

For each driver i , the paper defines two binary endogenous variables:

$$y_i = 1\{\text{driver } i \text{ bought comprehensive coverage (TR)}\}, \quad (1)$$

$$z_i = 1\{\text{driver } i \text{ had at least one at-fault accident in 1989}\}. \quad (2)$$

Thus:

- $y_i = 0$ means the driver purchased only RC coverage,
- $z_i = 0$ means no at-fault accident.

The paper does *not* distinguish among different deductibles within TR in the main analysis. The authors argue that deductible differences in their sample are small in expected-value terms relative

to the large gap between partial and full coverage.

2.6 Why the accident variable is defined carefully

The raw data record *claims*, not underlying accidents. This matters because a more comprehensive policy can make a driver more willing to report small losses even if the true accident process is unchanged.

To minimize this “accident versus claim” bias, the paper discards one-car accidents and keeps only multi-car accidents. The logic is that if two cars are involved, filing is much more likely regardless of the insured driver’s coverage choice.

Interpretation. This restriction is crucial. Without it, any positive relation between coverage and claims could be driven by reporting behavior rather than by hidden information about risk.

2.7 Observed characteristics

The insurer-side information set is rich. The data include the variables insurers themselves use for pricing: age, sex, profession, year of license, age and type of car, use of the car, and area of residence. In the main parametric beginner specification, the paper uses 55 exogenous variables plus a constant:

- sex (1 dummy),
- make of car (7),
- performance of car (5),
- type of use (3),
- type of area (4),
- age of driver (8),
- profession of driver (7),
- age of car (11),
- region (9).

For the fully nonparametric procedure, the authors collapse conditioning variables to six binary indicators chosen from insurer experience: sex, make of car, size of car, city use, age of driver, and age of car. This produces $2^6 = 64$ cells.

2.8 Exposure weighting

Drivers are observed for different fractions of the year, so each individual is weighted by w_i , the number of days in 1989 during which the individual was insured.

Interpretation. This is a simple but important measurement choice: someone insured for only part of the year mechanically has less exposure to accidents than someone insured for all 365 days.

2.9 Additional samples used later

The paper uses two additional samples beyond pure beginners.

- **More senior drivers.** Drivers who obtained their license in 1986 and had a bonus/malus no greater than 100 in 1989. There are 6,802 such drivers. This sample is meant to test whether private information emerges endogenously with driving experience.
- **Bonus-0.5 young drivers.** A subset of young drivers for whom the starting bonus coefficient equals 0.5 rather than the normal benchmark of 1. Roughly 19.5% of young drivers fall into this category. This institutional irregularity becomes the paper’s simple natural experiment for moral hazard.

3 Models and empirical specifications (all equations + interpretation)

3.1 Theory: what the paper is testing

The paper’s key theoretical prediction comes from competitive insurance models with exclusive contracts. Among observationally identical agents, the individuals who choose more comprehensive coverage should have higher accident probabilities if asymmetric information matters.

The empirical translation is not “do people with more coverage have more accidents on average?” but rather:

After conditioning on all observables known to insurers, are coverage choice and accident occurrence still positively related?

That is why the object of interest is *conditional dependence*, not raw correlation.

3.2 Pair of probits

The first parametric approach specifies two latent-index equations:

$$y_i = 1(X_i\beta + \varepsilon_i > 0), \tag{3}$$

$$z_i = 1(X_i\gamma + \eta_i > 0), \tag{4}$$

where ε_i and η_i are centered standard normal errors.

Interpretation.

- The first equation models the choice of comprehensive coverage.
- The second equation models the probability of an at-fault accident.
- Under symmetric information, once the observed variables X_i are conditioned on, the remaining latent components should be independent.

The two probits are first estimated separately. The generalized residuals are then computed. For

the coverage equation,

$$\hat{\varepsilon}_i = E(\varepsilon_i | y_i) = \frac{\phi(X_i\beta)}{\Phi(X_i\beta)}y_i - (1 - y_i)\frac{\phi(X_i\beta)}{\Phi(-X_i\beta)}, \quad (5)$$

with an analogous expression for $\hat{\eta}_i$.

The paper then forms the test statistic

$$W = \frac{(\sum_{i=1}^n w_i \hat{\varepsilon}_i \hat{\eta}_i)^2}{\sum_{i=1}^n w_i^2 \hat{\varepsilon}_i^2 \hat{\eta}_i^2}, \quad (6)$$

which is asymptotically $\chi^2(1)$ under the null of conditional independence.

Interpretation. If residual coverage choice and residual accident risk comove after conditioning on insurer observables, then the data contain evidence of asymmetric information.

3.3 Bivariate probit

The second parametric approach estimates a bivariate probit in which the latent errors remain standard normal but are allowed to be correlated:

$$\text{corr}(\varepsilon_i, \eta_i) = \rho. \quad (7)$$

The null hypothesis of symmetric information is:

$$H_0 : \rho = 0. \quad (8)$$

Interpretation. The bivariate probit is a more direct and potentially more efficient way to estimate the latent correlation between coverage choice and accident occurrence.

3.4 Fully nonparametric χ^2 independence test

Because the probit equations still impose linear latent indices and normal errors, the authors also build a nonparametric robustness check.

Take a vector x_i of m binary conditioning variables. These define 2^m cells in which people are identical on the selected observables. Inside each cell, the paper constructs the following 2×2 table:

Population	$z_i = 0$	$z_i = 1$
$y_i = 0$	N_{00}	N_{01}
$y_i = 1$	N_{10}	N_{11}

Within each cell, the independence statistic is

$$T = \sum_{j,k=0,1} \frac{[N_{jk} - (N_{j.}N_{.k}/N_{..})]^2}{N_{jk}}, \quad (9)$$

which is asymptotically $\chi^2(1)$ under conditional independence.

The paper then aggregates information across the $M = 2^m$ cells in three ways:

- a Kolmogorov–Smirnov statistic based on the empirical cdf of the cell-level T ’s,
- the number of cells in which $T > 3.84$ (the 5% critical value of $\chi^2(1)$),
- the sum of all cell statistics,

$$S = \sum_{\text{cells}} T, \tag{10}$$

which is asymptotically $\chi^2(M)$ under the null.

Interpretation. This procedure asks the same question as the probits but in a much less parametric way. If all three approaches tell the same story, the conclusion is much more credible.

3.5 Why the paper uses several methods

The empirical strategy is deliberately redundant.

- The separate probits plus generalized residuals provide a simple omnibus test.
- The bivariate probit gives a direct estimate and confidence interval for the latent correlation.
- The nonparametric test protects against misspecified functional form.

This is one of the paper’s strongest features: the conclusion is not hanging on one estimator.

3.6 Sample-specific implementation

For pure beginners (drivers licensed in 1988), the paper uses the 55-variable X_i in the parametric models. For the nonparametric test, it reduces the conditioning set to six binary variables, creating 64 cells. This is a practical trade-off between richer conditioning and enough observations per cell for asymptotics to make sense.

3.7 Specification for the experienced-driver test

The test for more senior drivers keeps the same three empirical procedures but changes the sample. The idea is to ask whether hidden information appears only after drivers accumulate personal experience. The authors select drivers licensed in 1986 who still have bonus/malus ≤ 100 in 1989 and re-run all three procedures on that subsample.

Interpretation. If drivers learn privately about their own risk faster than insurers do, experienced drivers should show stronger dependence between coverage and accidents than true beginners.

3.8 Specific moral-hazard test using the bonus coefficient

The paper’s bonus-0.5 exercise is not a full structural model of moral hazard, but it is a clever reduced-form test. Let $B_i = 1$ if the young driver starts with bonus coefficient 0.5. The paper adds B_i to the coverage and accident probits.

The logic is:

- if lower out-of-pocket accident cost weakens incentives, then B_i should increase accident probability under moral hazard,
- while it should also raise demand for comprehensive coverage because insurance becomes effectively cheaper.

The first sign is what matters for the moral-hazard diagnosis.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline result

The paper has no external instrument in the usual IV sense. Identification comes from a conditional-independence logic:

If insurers and drivers observe the same relevant risk information, then once the insurer's covariates are controlled for, residual coverage choice and residual accident occurrence should be independent.

So the baseline estimate is identified from within-observable-type comparisons, not from cross-market or cross-time experiments.

4.2 Why concentrating on beginners helps

The most important design choice is to begin with drivers who obtained a license only recently. This matters because the biggest threat to a clean adverse-selection test is the driver's past record. Insurers see past accidents and use them in pricing, so if the econometrician omits driving history, some public information is incorrectly treated as private information.

Using beginners sharply weakens that problem. It also avoids the need to treat driving experience as exogenous, which it is not.

4.3 Why reverse causality is a real issue here

A central conceptual problem is that the same reduced-form correlation can come from two different directions:

- **Adverse selection:** risky drivers choose more coverage.
- **Moral hazard:** more coverage induces less prevention, so accidents rise after the contract is chosen.

The paper is fully explicit that its main static test cannot, by itself, distinguish those stories. That is why the bonus-0.5 institutional feature is useful: it changes incentives in a way that helps isolate moral hazard.

4.4 Why dropping one-car accidents matters

Without additional care, the outcome variable would confound accidents with the decision to file a claim. A more comprehensive policy makes reporting more attractive. By discarding one-car accidents, the authors try to keep the dependent variable closer to actual accident risk.

This is not a perfect solution, but it is a serious attempt to remove an obvious false-positive channel.

4.5 Why multiple methods improve credibility

The paper's identification is strengthened by triangulation.

- The generalized-residual test says the latent dependence is tiny.
- The bivariate probit says the latent correlation ρ is statistically indistinguishable from zero.
- The nonparametric cell tests also fail to reject dependence.

If all three methods were sensitive to different modeling assumptions yet produced the same answer, it becomes much harder to claim that the main result is a parametric artifact.

4.6 Why the critique of Puelz and Snow matters

A large part of the paper's credibility section is a critique of earlier positive findings. The authors argue that Puelz and Snow's evidence can be spurious for four reasons:

- using realized accidents as a proxy for latent accident risk introduces measurement error,
- relying on highly restricted linear functional forms can confound omitted nonlinearities with adverse selection,
- omitting driving history mistakes insurer-observed information for private information,
- mixing drivers with very different seniority creates heteroskedasticity and further misspecification.

Interpretation. The paper's negative result is not simply "we find zero." It is "we find zero after fixing what we think generated earlier false positives."

4.7 What the paper can and cannot distinguish

The paper is careful on scope.

- It tests for **risk-related** asymmetric information showing up as a positive conditional correlation between coverage and accidents.
- It does *not* fully rule out all forms of hidden information.
- In particular, hidden heterogeneity in preferences or risk aversion could still matter without generating the same positive coverage-accident correlation.

This is the paper's "cherry-picking" discussion: more risk-averse individuals may buy more insurance and drive more carefully, possibly producing zero or even negative correlation.

TABLE 1
TESTING INDEPENDENCE

Population	$z_i = 0$	$z_i = 1$
$y_i = 0$	N_{00}	N_{01}
$y_i = 1$	N_{10}	N_{11}

4.8 Limitations

Several limitations remain.

- The data are cross-sectional, so dynamic testing of learning and moral hazard is limited.
- The paper cannot observe all relevant private variables directly; it argues instead that any omitted variable would have to create an exact offsetting cancellation to overturn the negative result.
- The sample is specific to French automobile insurance and to the regulatory environment of that market.
- The accident-versus-claim correction requires excluding one-car accidents, so the paper may miss asymmetric information that matters particularly for those events.

4.9 Why the authors think the negative finding is robust

A memorable argument in the conclusion is the idea of an *asymmetric danger*. Misspecification can easily produce false positives — apparent adverse selection where none exists. But it is harder to see how misspecification alone would hide a strong true dependence. The authors therefore argue that a carefully obtained negative result may be more robust than a positive one.

5 Tables and figures

5.1 Table 1: the basic independence structure

Read:

- The paper’s only formal table is a simple 2×2 contingency layout crossing coverage choice and accident occurrence.
- Each cell counts how many individuals in a conditioning cell buy RC versus TR and then do or do not have an at-fault accident.

Economic message. The paper is deliberately simple. It reduces the empirical test to the most basic implication of hidden risk information: conditional dependence between coverage and accidents.

5.2 Main reported results for beginners

Read:

- The beginner sample consists of 6,333 drivers licensed in 1988.
- In the separate-probit residual test, the statistic is $W = 0.46$, far too small to reject independence.
- In the bivariate probit, the estimated latent correlation is $\hat{\rho} = -0.029$ with standard error 0.049 and 95% confidence interval $[-0.125, 0.067]$.
- In the nonparametric procedure, the Kolmogorov–Smirnov statistic is $K = 0.63$, well below the 5% critical value of 1.35.
- Only 5 of the 64 cells reject independence, corresponding to about a 15% binomial p -value.
- The sum statistic is $S = 69.3$, below the 5% critical value of 87.5 for $\chi^2(64)$.

Economic message. Every method says the same thing: among beginners, comprehensive coverage is not associated with hidden residual risk once the insurer’s information is controlled for.

5.3 Results for more senior drivers

Read:

- The experienced-driver subsample contains 6,802 drivers licensed in 1986 with bonus/malus no greater than 100 in 1989.
- The generalized-residual statistic is even smaller than before: $W = 0.25$.
- The bivariate probit estimate is $\hat{\rho} = 0.024$ with standard error 0.057 and 95% confidence interval $[-0.088, 0.136]$.
- The nonparametric test gives only 1 rejection out of 64 cells.
- The sum statistic is $S = 59.3$ and the Kolmogorov–Smirnov statistic is $K = 0.82$.

Economic message. The absence of dependence is not just a “true beginners know nothing” result. Even after some driving experience, the paper still finds no sign that drivers possess extra private information about their risk relative to insurers.

5.4 Bonus-0.5 moral-hazard test

Read:

- About 19.5% of young drivers start with the minimum bonus coefficient of 0.5 rather than the standard coefficient of 1.
- This favorable coefficient makes comprehensive coverage more attractive, and indeed the coverage probit shows a positive and significant coefficient on the bonus-0.5 indicator.
- But the accident probit shows a *negative* and significant coefficient on the same indicator.
- The residual-independence test remains very small.

Economic message. If weaker incentives under better insurance terms were producing more accidents, the accident effect should be positive. It is not. This is strong evidence against the idea that the main findings are hiding moral hazard.

5.5 What is visually absent is also informative

The paper has no figures and almost no large result tables. Most of the quantitative evidence is reported directly in the text as test statistics, standard errors, confidence intervals, and critical values.

Economic message. This reflects the paper’s design philosophy: it is not trying to estimate a rich structural model of insurance demand. It is asking a tightly focused yes-or-no question and answering it with several complementary statistical tests.

6 Short summary

Chiappori and Salanié propose a simple empirical test of asymmetric information in competitive insurance markets: after conditioning on everything insurers observe, people who choose more comprehensive coverage should also be riskier *ex post*. They apply this idea to French automobile insurance using several parametric and nonparametric methods. Focusing first on beginners and then on somewhat more experienced drivers, they find essentially no residual relation between coverage and accidents. They also exploit an institutional irregularity in the French bonus/malus system to test moral hazard and again find no support for it. The paper’s main substantive conclusion is that risk-related asymmetric information appears to be negligible in this market once observables are handled carefully. Its deeper methodological message is that earlier positive findings may have been driven by misspecification, omitted observables, or a failure to separate claims from accidents.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that, in the French automobile insurance market studied here, there is no detectable evidence of risk-related asymmetric information once the econometrician conditions on the rich set of observables available to insurers. This conclusion survives separate probits, bivariate probit estimation, and a fully nonparametric independence test. The paper also fails to find evidence that moral hazard is driving the correlation one might expect between broader coverage and accidents.

7.2 Economic intuition

The natural intuition behind adverse selection is simple: people know more about their own risk than insurers, so the riskiest drivers buy the most insurance. Chiappori and Salanié show that this intuition can fail in practice when insurers already observe very informative proxies for risk and when the econometrician conditions on them properly. In that case, the remaining private-information problem may be tiny. The paper also reminds us that insurance choices can be shaped by preferences,

family arrangements, reporting behavior, and regulation, so raw positive correlations are not enough.

7.3 Takeaway

This paper is influential because it changes the empirical standard. Instead of asking whether insured people with more coverage look riskier on average, it asks whether they still look riskier after conditioning on what insurers know. In this application, the answer is no. The broader lesson is methodological as much as substantive: in insurance data, rich conditioning and robustness across methods matter enormously, and positive evidence of adverse selection is much easier to generate spuriously than negative evidence is to hide.